

range



Agenda

- ① What are iterables?
- ② various iterables
- ③ range
- ④ Accessing range elements
- ⑤ Practice

What are iterables?

An iterable object is something that contains a countable number of values.

You can traverse through all the values from beginning to the end.

Technically, in Python, an iterator is an object, which implements the iterator protocol, which consist of methods `--iter--()` and `--next--()`

Various Iterables

- range
- list
- tuple
- str
- bytes
- bytearray
- set
- frozenset
- dict

range

- range is a class
- range is immutable sequence
- range can contain only int type values
- range contains sequence of integers with common difference (Arithmetic Progression)
- range elements are indexed

How to create range object?

$x = \text{range}(\text{beg}, \text{end}, \text{step})$

Example

$\text{range}(1, 10, 1)$ 1 2 3 4 5 6 7 8 9
 ↑ ↑ ←
 inclusive Exclusive common gap

$\text{range}(2, 8, 2)$ 2 4 6

$\text{range}(10, 3, -2)$ 10 8 6 4

Other ways to create range object

$x = \text{range}(\text{beg}, \text{end}, \text{step})$

① $\text{range}(\text{end})$ $\text{beg} = 0$ $\text{step} = 1$

② $\text{range}(\text{beg}, \text{end})$ $\text{step} = 1$

indexing

$x = \text{range}(1, 6, 1)$

-5	-4	-3	-2	-1
0	1	2	3	4
1	2	3	4	5

$x[0] \Rightarrow 1$

$x[1] \Rightarrow 2$

$x[4] \Rightarrow 5$

Positive and negative indexing

$x[-1] \Rightarrow 5$

$x[-2] \Rightarrow 4$

$x[-4] \Rightarrow 2$

$x[-5] \Rightarrow 1$

Accessing range elements.

for e in r:

Code

e is an element of range

Write a program to print first n natural numbers. [use range and for]

$n=6$

1 2 3 4 5 6

`range(1, n+1, 1)`

Write a program to print
Squares of first n natural numbers
[use range and for]

$n=5$

1^2	2^2	3^2	4^2	5^2
1	4	9	16	25

Write a program to print first n even natural numbers in reverse order. [use range and for]

$n = 5$

1 (2) 3 (4) 5 (6) 7 (8) 9 (10) 11 (12) ...
1 2 3 4 5
10 8 6 4 2

range(10, 1, -2)
↑
 $2 * n$

Write a program to calculate
Sum of first n multiples of
 x . [use range and for]

$$n = 5$$

$$x = 3$$

First 5 multiples of 3

$$3 + 6 + 9 + 12 + 15$$

$$= \underline{\underline{45}}$$

range(1, $n+1$, 1)

1	2	3	4	5
x	$2x$	$3x$	$4x$	$5x$

$$x = 3$$

$$n = 5$$

$$S = 0$$

$$S = S + x * e$$

$$\text{range}(1, n+1, 1)$$

e 1 2 3 4 5

S	x * e		S
0	+ 3	→	3
3	+ 6	→	9
9	+ 9	→	18
18	+ 12	→	30
30	+ 15	→	45